



ÉCOLE POLYTECHNIQUE FÉDÉRALE DE LAUSANNE

**Teleoperation for Anthropomorphic Robotic
Manipulation using RL-Based Robust Retargeting**

MASTER THESIS

Author:

Jeffrey YU, 371327

CREATE Lab, Prof. Josie Hughes

Supervisor: Cheng Pan

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1 Introduction

Teleoperation for robotic dexterous manipulation is a very difficult task. Some reasons include Operator Hand shape differences Manipulator hand shape differences We see that there are approaches to perform manipulation using RL, as the policies can learn how to maximize the contact forces, etc. with the object. However, many of these approaches use object-specific reward functions or train a single policy for a single demonstration. Using a policy for teleoperation for single task would therefore be infeasible as the policy is fitted to one instance.

2 Methods

2.1 RL Formulation

The manipulation task is formulated using a reinforcement learning paradigm. The tasks are in the family of dexterous bimanual manipulation demonstrations, where the goal is to drive a pair of dexterous hands $\mathbf{d} = \{d_l, d_r\}$ to perform a certain manipulation task with two objects in a collaborative manner, for example in this paper the capping of an alcohol bottle. The policy takes as inputs a reference trajectory from the human: $\mathbf{h} = hl, hr$ and the objects: $\mathbf{o} = \{o_l, o_r\}$.

Describe the training method, 2-stage policy, policy inputs,

2.1.1 Training Method

Building upon the Maniptrans paper, we use the pretrained imitator model, and train the residual model from scratch. The Describe using a curriculum to adjust the gravity, friction, and early termination threshold. Initially, the gravity

Generation of Multiple Trajectories insert image of modified trajectory here

Ontop of collected trajectories, we generate trajectories to increase the model generalizability. We apply affine transformations to the hand and/or objects across the entire trajectory.

Joint Position Noise

Since we are using inherently noisy data collection methods, joint noise was added ontop of the wrist and finger cartesian positions. The Apple Vision had a RMSE of around 8-12 mm with respect to the motion capture.

$$\tilde{p}_{\text{wrist}} = p_{\text{wrist}} + \epsilon_{\text{wrist}}, \quad \epsilon_{\text{wrist}} \sim \mathcal{U}(-\sigma, \sigma)^3, \quad (1)$$

$$\tilde{j}_k = j_k + \epsilon_k, \quad \epsilon_k \sim \mathcal{U}(-\sigma, \sigma), \quad k \in \mathcal{J} \quad (2)$$

2.2 Experimental Setup

insert image of block diagram demonstrating how data flows

Discuss the hardware used: MOCAP AVP

To collect data and perform teleoperation, we use two systems, the Apple Vision Pro (AVP) and the OptiTrack running at 60 Hz. Since the two systems have different coordinate systems,

we perform a calibration using a tracker on the tip of the index finger of the right hand. By moving the hand around in the workspace of the AVP and the OptiTrack, we have a set of two points that we use the Uemama algorithm to map from the AVP to Optitrack.

We train the residual module R with the actor-critic PPO algorithm [1], using a training horizon of 32 frames, a mini-batch size of 1024, and a discount factor of $\gamma = 0.99$. Optimization uses Adam [2] with an initial learning rate of 5×10^{-4} and a decay scheduler. All simulation is carried out in Isaac Gym [3] with a time step of 1/60 s. Training is performed on an HPC cluster equipped with NVIDIA V100 and A100 GPUs, while evaluation, visualisation, and teleoperation are run on a single NVIDIA RTX 3080 GPU.

Discuss the software pipeline: Real time streaming to the IsaacGym simulator at near 60 fps

2.3 Increasing generalizability the model

3 Results

The generalizability of the model was verified using LOO training

ManipTrans Baseline

Policy trained on a single demo, running other demonstration performance

Policy trained on multiple demos running other demonstrations

4 Discussion

5 References

- [1] John Schulman et al. “Proximal Policy Optimization Algorithms”. In: *arXiv preprint arXiv:1707.06347* (2017). arXiv: [1707.06347 \[cs.LG\]](#). URL: <https://arxiv.org/abs/1707.06347>.
- [2] Diederik P. Kingma and Jimmy Ba. “Adam: A Method for Stochastic Optimization”. In: *International Conference on Learning Representations (ICLR)*. 2015. arXiv: [1412.6980 \[cs.LG\]](#). URL: <https://arxiv.org/abs/1412.6980>.
- [3] Viktor Makoviychuk et al. “Isaac Gym: High Performance GPU-Based Physics Simulation for Robot Learning”. In: *arXiv preprint arXiv:2108.10470* (2021). arXiv: [2108.10470 \[cs.R0\]](#). URL: <https://arxiv.org/abs/2108.10470>.

6 Appendix